AE 196: ENGINEERING WORKSHOP PRACTICE (1,3,2)

DR. K. A. AIKINS

DR. E. A. ASANTE

Course Description

- An Introduction to Technical Report Writing
- Study of Hand Tools, Fitting Tools, & Power tools.
- Machine Tool Work using Working Drawings & Metrology:
  - Turning,
  - Milling and
  - Shaping.

Course Description

- Machine Building Exercises.
- Welding: Gas & Electric Exercises.
- Sheet Metal Work exercises.

TECHNICAL REPORT WRITING

Elements of Good Technical Communication

- **Clear**
  - Unambiguous
  - Exact meaning conveyed
  - Symbols fully defined
  - Figures and tables easily understood

Elements of Good Technical Communication

- **Concise**
  - Omission of irrelevant data
  - Try to answer the Who, What, Why, Where and the How
  - Be careful not to oversimplify

Elements of Good Technical Communication

- **Concise**
  - Omission of irrelevant data
  - Try to answer the Who, What, Why, Where and the How
  - Be careful not to oversimplify

Elements of Good Technical Communication

- **Appropriate writing Style**
  - Active vs. Passive Voice (e.g., “The length of the bar was measured” instead of “I measured the length of the bar”)
  - Use words to convey the appropriate meaning
  - Avoid the use of nonstandard jargons
  - Define jargon if necessary

Elements of Good Technical Communication

- **Guide your reader**
  - Tell readers what you plan to tell them (Introduction).
  - Then tell them (main text).
  - Finally tell them what you told them (Summary of Results or Conclusions).

Elements of Good Technical Communication

- **Guide your Reader**
  - State your purpose clearly
  - Get to the point. Don’t beat about the bush
  - Emphasize major ideas

Elements of Good Technical Communication

- Present the information from your work, both real world and theoretical, or your design/ drawings.
- Organise information logically under appropriate headings.
- Convey information in the most effective way for communication:
  - use figures and tables (a picture speaks more than a thousand words, but not excessive use of pictures)
  - You can use bulleted or numbered lists

Parts of a Technical Report

- Title page
- Abstract or summary
- Acknowledgements
- Table of Contents
- List of figures, graphs and tables

Parts of a Technical Report

- Introduction
- Body
- Conclusion
- References
- Appendix (ces)

Title Page

- This page gives:
  - Name of the university, faculty, college and department.
  - Course code and name
  - Title of the report
  - Authors' name and Index number
  - the date of submission.

Example of Title Page

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
KUMASI, GHANA

COLLEGE OF ENGINEERING

FACULTY OF MECHANICAL AND CHEMICAL ENGINEERING

DEPARTMENT OF AGRICULTURAL ENGINEERING

AE 196 ENGINEERING TECHNOLOGY II

METAL SHEET FABRICATION

[KNUST LOGO]

NAME: KWEKU ANANSE

INDEX NO.: 0000000000

DATE: 1ST JANUARY, 1900.

TABLE OF CONTENTS

DECLARATION i
ABSTRACT ii
DEDICATION iii
ACKNOWLEDGEMENTS iv
TABLE OF CONTENTS v
LIST OF TABLES vii
LIST OF FIGURES vii

1. INTRODUCTION 1
   1.1 Background to the Study 1
   1.2 Significance of the Study 2
   1.3 Aim and Objectives 3
2. LITERATURE REVIEW 4
   2.1 Weeds 4
   2.2 Weed Control 4
   2.3 Tractor Operated Weeding Tools 7
      2.3.1 Rotary hoe 7
      2.3.2 Tine Weeder 7
3. MATERALS AND METHODS 10
   3.1 Design Considerations 10
   3.3 Soil Force Prediction 11
      3.3.1 Narrow Tine Force Prediction 11
4. RESULTS AND DISCUSSION 17
5. CONCLUSIONS AND SUUGESTIONS FOR FURTHER WORK 29
REFERENCES 30
APPENDIX 33

List of Tables and Figures

- Figures include pictures, graphs and charts.
- The list should include the table or figure number, the exact title and the page where it is found.
- Tables and figures should stand on their own, i.e., understandable without reference to the text
- but they need to be referred to in the text.

List of Tables and Figures

LIST OF TABLES

Table 3.1: Values of parameters used for calculating Draught & Vertical Forces 14
Table 4.1: Results of Design Calculations and Specifications..... 20

LIST OF FIGURES

Fig. 2.1: Cutlass and Hoe 5
Fig. 2.2: Manually Operated Row Crop Weeders..... 6
Fig. 2.3: An Animal Drawn Weeder 6
Fig. 2.4: A six-row rotary hoe..... 8
Fig. 2.5: A tine weeder..... 8

Abstract

- It is the summary of the report.
- It communicates the contents of report.
- It includes purpose (objectives), methods, results, conclusions, and recommendations.
- It highlights essential points of the report.
- It is short--from a paragraph to a page or two, depending upon the length of the report (10% or less of the report)

A Good Abstract:

- uses one or more well-developed paragraphs, which are unified, coherent, concise, and able to stand alone
- uses an introduction-body-conclusion structure in which the parts of the report are discussed in order:
  - Aim/objectives, methods/procedure, findings, conclusions, recommendations
- follows strictly the chronology of the report
- adds no new information but simply summarizes the report
- is intelligible to a wide audience

Introduction

- Background to the study, experiment, etc.
- Statement of purpose (Aim(s) and Objectives)
- Scope/context/boundaries
- **Do not show figures, tables and bulleted/numbered points in the introduction**
- Figures, tables and bulleted/numbered points are shown in the literature review.
- In a short report, the Introduction incorporates the Literature Review.
- In that case, figures, tables and bulleted/numbered points can be shown.

Sample Introduction

1. Introduction

A dual carriageway bridge with two traffic lanes in each direction is to be constructed on the Calder Freeway crossing Slaty Creek in the Shire of Macedon Ranges in Victoria. The bridge is to span 125 metres between man-made compacted fill embankments, and is approximately 15 metres above the river surface, with a grade of 0.056 m/m.

This report presents two possible concept designs for the bridge. In evaluating these designs, the following criteria are considered: construction method, construction and maintenance costs, possible disruption to traffic during construction, the durability and the aesthetics of the bridge.

The two conceptual designs are presented in the form of sketches of the elevations and cross-sections of the structures.

Why Literature Review?

- You should know what has already been done in the line of study and the significance of other reported research bordering on the study.
- This will prevent re-inventing the wheel!
- It shows approaches that can be followed for successful project.
- Record the main outlines and credit the various sources (referencing within the text).
- Example: “According to Agodzo (2003a) it is indicated …”

Materials and Methods

- Experiment and Analysis Description
  - Apparatus Sections
  - Materials Section
  - Test Procedure Section
- Sufficient information must be given to allow others to repeat the work if necessary, but avoid irrelevant details

Results and Discussion

- Present results in tables and figures.
- Allow the tables and figures to speak for themselves, but draw attention to the most important features
- Report negative results but inconclusive experiments can be omitted
- The results must be presented in terms of progress towards the scientific objectives
- Describe important features of graphs in words under the results

Fig. 1 illustrates the educational level of the tractor operators who participated in the study. Out of the 61 tractor operators, 47.5% had no formal education.

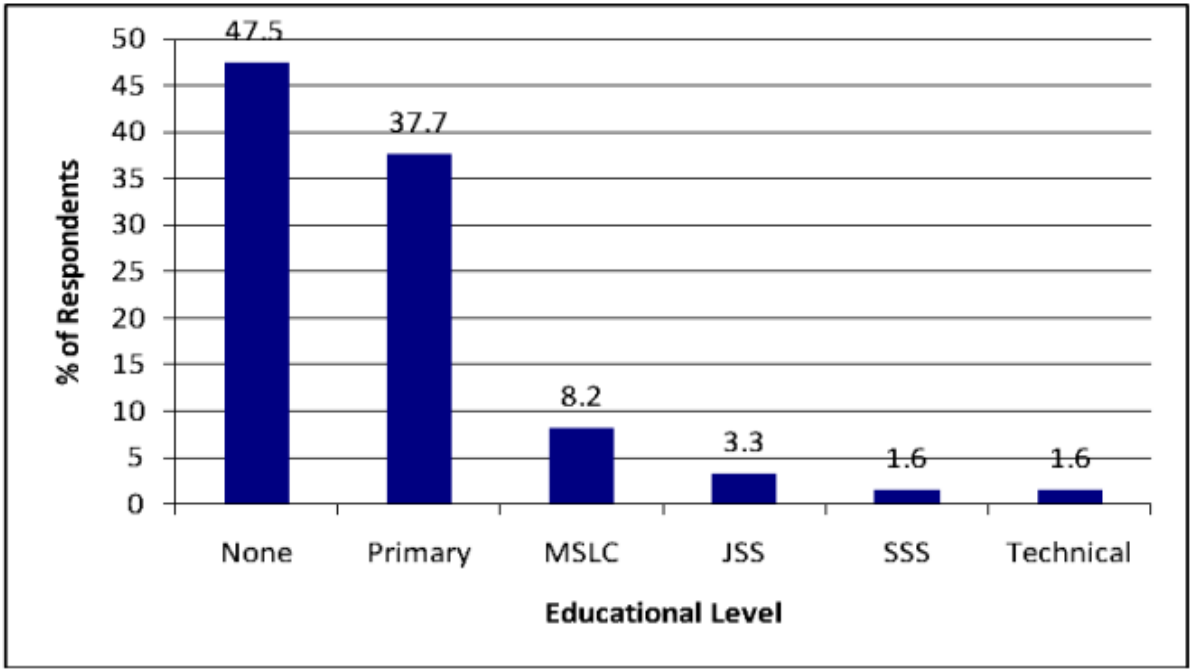


Fig. 1: Educational Level of Tractor Operators

Approximately 37.7% of the tractor operators had between one and six years of basic school primary education while about 11.5% of the respondents had middle school leaving certificate (MSLC: 8.2%) or Junior Secondary School (JSS: 3.3%) level education. Only 3.2% of the tractor operators had education above the basic school level [i.e. Senior

Tables and Figures

- Give each figure and table a number and title.
- Refer to each figure and table in the text of the report.
- The title of a table goes above the table, while the title of a figure goes below the figure

Tables and Figures

Table 1 Turning volume of pedal cycles

	(1)	(2)	(3)	(4)	(5)	(6)
8:00 - 8:15am	0	0	1	0	1	0
8:15 - 8:30am	0	0	1	1	0	0
8:30 - 8:45am	0	0	3	1	0	0
8:45 - 9:00am	0	0	2	3	1	0
Total Volume	0	0	7	5	2	0

Tables and Figures

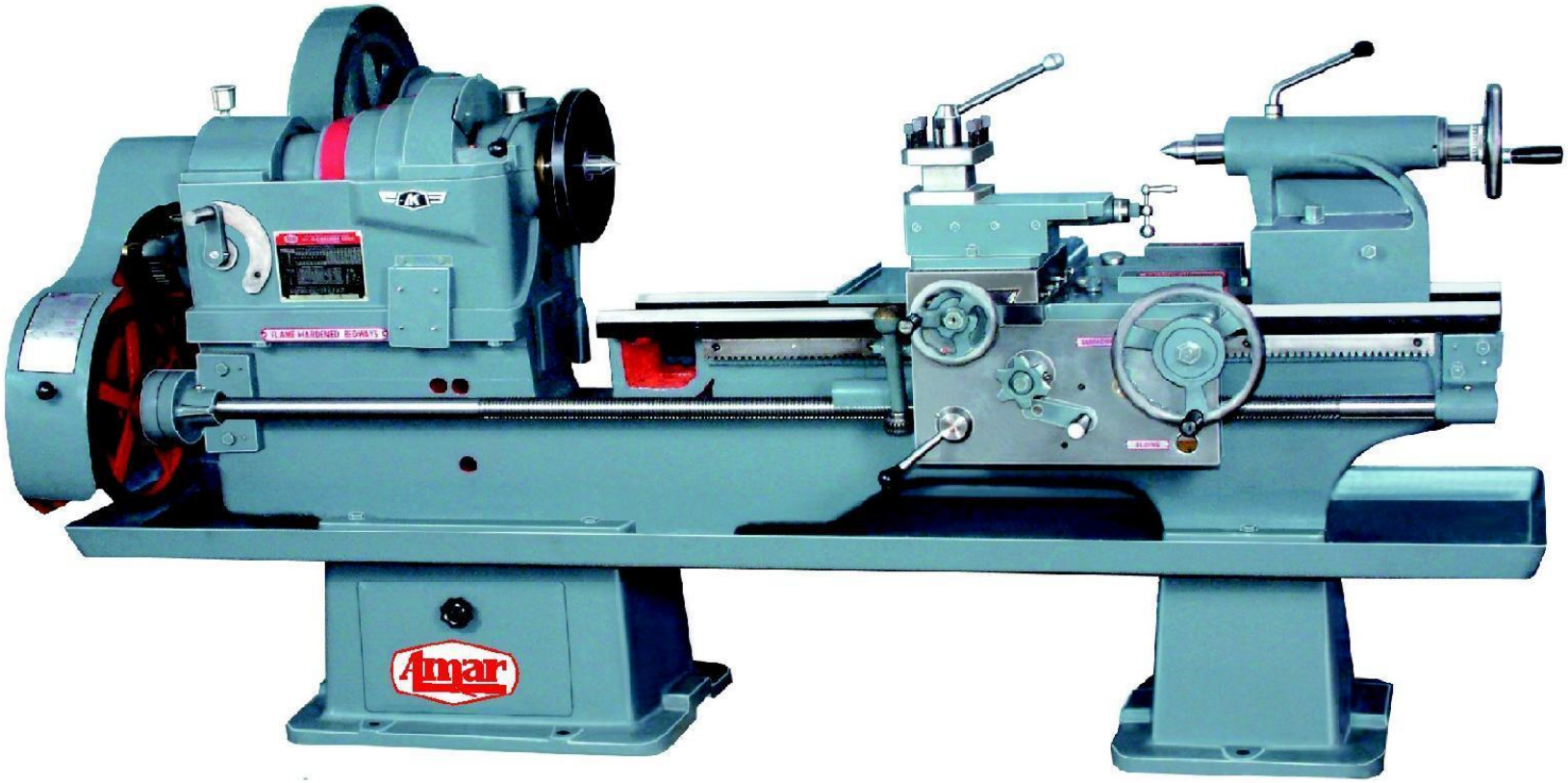


Figure 1: Centre lathe

Conclusion

- Summary of Results
- Conclusions
- Concluding Remarks

Referencing

- References serve as a guide to any reader who wants to refer to previous work on the subject.
- It is also meant to show appreciation for the invaluable work of other researchers.
- Failure to credit the work of others is regarded as:
  1. Carelessness on the part of the author
  2. Lack of courtesy and even more seriously
  3. Plagiarism, in which case the student would be failed

Referencing

- A reference to some document in the write-up should simply have the surname of the author and the year of publication
- If there are two authors, both names are given before the date
- If there are more than two authors, the surname of the first author is given, followed by "*et al*" in italics and then the year
- If one author has published more than one cited document in the same year, these are distinguished by adding lower case letters after the year.

Referencing

Referencing/ Citation in the text

Within a sentence

- "According to Agodzo (2003a) it is indicated …"
- "Equations developed by Bart-Plange *et al* (2002) can be used to …"
- "Kyei-Baffour (2004) outlines the principle for the design of simulators…"

At the end of a sentence

- You can pass this course easily (Aikins, 2015).

List of References

- **Journal**: Bart-Plange, A. and Baryeh, E. A. (2003). The physical properties of Category B cocoa beans. Journal of Food Engineering. 20(1): 219-227
- **Book**: Morgan, R. P. C. (1986). Soil erosion and conservation. Longman, U.K.
- **Contribution to a book**: Kyei-Baffour, N., Bobobee, E. Y. H. and Koram, S. A. (1994). The effects of tillage and irrigation practices on soils in Ghana: A case study. In: L. O. Fresco, L. Stroosnijder, J. Bouma and H. van Keulen, eds. The future of the land. Willey, Sussex. 363-364
- **Technical Report**: Agodzo, S. K., Huibers, F. P., Chenini, F., van Lier, J. B. and Duran, A. (2003). Use of wastewater in irrigated agriculture. Country studies from Bolivia, Ghana and Tunisia. Volume 2: Ghana. Wageningen WUR (WF4-Wastewater)-ISBN 90-6754-704-2

List of References

References

Agco Allis (2011). Tractor, Service and Operator's Manuals. Electronic document. Retrieved from: http://www.tractormanual.info/. Accessed 3 May 2011.

Bhutta, M.S., Tanveer, T. & Awan, H.M. (1997). Technical Skill of Tractor Operator: A Case Study in Multan, Pakistan. *Agricultural Mechanisation in Asia, Africa and Latin America*, 28 (1): pp. 18–22.

Davies, D.B., Eagle, D.J. & Finney, J.B. (1993). *Soil Management*, Ipswich, Farming Press.

MINITAB Inc. (2007). *MINITAB Statistical Software Release 15 for Windows*. Pennsylvania, Minitab Inc., State College.

Appendix (ces)

- Any material which does not fit easily into the mainstream of the argument, but is relevant to the work, should be put into an appendix

- Detailed results of experimental work can be included here so that anyone questioning the reasoning or wishing to pursue it further may have access to the material on which it is based